

SEPARABLE EQUATIONS

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KVB

⊕ Easiest to solve!

- Include non-linear equations
- Have a special feature that makes them approachable.

$$\frac{dy}{dt} = \frac{g(t)}{f(y)} \quad \text{or} \quad \frac{dy}{dt} = f(y)g(t)$$

{ functions of t & y have separated on the right hand side }

Counter example:

$$\frac{dy}{dt} = y + t$$

- This is a simple ODE
- But is not a separable ODE

→ Here, we can do an integral of

- (i) $f(y)$ and
- (ii) $g(t)$; and we are in business!

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KVB

Let's just take a very simple

examples:

$$\boxed{\frac{dy}{dt} = \frac{g(t)}{f(y)}} \quad \leftarrow \text{Big picture}$$

$$\textcircled{1} \quad \underline{\text{Say, } f(y) = 1} \quad \boxed{\frac{dy}{dt} = g(t)}$$

$$\Rightarrow dy = g(t) dt$$

$$\Rightarrow \int dy = \int g(t) dt + C$$

$$\Rightarrow \boxed{y = \int g(t) dt + C} \quad \leftarrow \text{Arbitrary constant}$$

OK

$$\boxed{y = \int_0^t g(s) ds} \quad \leftarrow \text{Definite integral}$$

Dummy variable (s)

$$\textcircled{2} \quad \underline{\text{Suppose, } g(t) = 1} \quad \boxed{\frac{dy}{dt} = \frac{1}{f(y)}}$$

$$\Rightarrow f(y) dy = dt$$

$$\Rightarrow \int f(y) dy = \int dt + C$$

$$\Rightarrow \boxed{\int f(y) dy = t + C} \quad \leftarrow \text{Arbitrary constant}$$

[Integral we have to do]

This is the
minimal
amount of work
we need to do

(3) Case where we have both $g(t)$ & $f(y)$:

$$\frac{dy}{dt} = \frac{g(t)}{f(y)}$$

$$\Rightarrow f(y) dy = g(t) dt$$

$$\Rightarrow \int f(y) dy = \int g(t) dt + C$$

Integration
we have to do

OR

$$\int_{y(0)}^{y(t)} f(y) dy = \int_0^t g(s) ds$$

Either ways, we have integrate (on both sides)
and solve for y !

(4)

$$\frac{dy}{dt} = \frac{t}{y}$$

Solve for y

$$\Rightarrow y dy = t dt$$

INDEFINITE INTEGRAL:

$$\Rightarrow \int y dy = \int t dt + C$$

$$\Rightarrow \frac{y^2}{2} = \frac{t^2}{2} + C$$

$$\Rightarrow y = \sqrt{C^2 + t^2}$$

DEFINITE

$$\int_{y(0)}^{y(t)} y dy = \int_0^t t dt$$

$$\frac{[y(t)]^2}{2} - \frac{[y(0)]^2}{2} = \frac{t^2}{2} - \frac{0^2}{2}$$

$$\Rightarrow y(t) = \sqrt{y(0)^2 + t^2}$$

⑤

LOGISTIC EQUATION

Recall
we can use $z = \frac{1}{y}$ ^{KVB}

$$\frac{dy}{dt} = y - y^2$$

$$\Rightarrow \frac{dy}{y - y^2} = dt$$

$$\Rightarrow \int \frac{dy}{y - y^2} = \int dt + C$$

$$\Rightarrow \int \frac{dy}{y - y^2} = t + C$$

INTEGRATING
WHEN THERE'S A
POLYNOMIAL IN
THE DENOMINATOR



There's a method
of integration



PARTIAL FRACTIONS

Consider

$$\left[\frac{1}{y - y^2} = \frac{1}{y(1-y)} = \frac{A}{y} + \frac{B}{1-y} \right]$$

$$\Rightarrow \int \frac{A dy}{y} + \int \frac{B dy}{1-y} = t + C$$

$$\Rightarrow \int \frac{dy}{y} + \int \frac{dy}{1-y} = t + C$$

$$\Rightarrow \ln(y) - \ln(1-y) = t + C \quad \left[\begin{array}{l} A=1 \\ B=1 \end{array} \right]$$

$$\Rightarrow \ln\left(\frac{y}{1-y}\right) = t + C$$

$$\Rightarrow \frac{y}{1-y} = Ke^t \Rightarrow y = \frac{Ke^t}{1 + Ke^t}$$

$$\Rightarrow y(1 + Ke^t) = Ke^t$$

$$\Rightarrow y = \frac{Ke^t}{1 + Ke^t} \Rightarrow y(t) = \frac{1}{1 + Ke^{-t}}$$

$$\frac{1}{y(1-y)} = \frac{A(1-y) + By}{y(1-y)}$$

$$\Rightarrow \frac{1}{y(1-y)} = \frac{A - Ay + By}{y(1-y)}$$

$$\Rightarrow \frac{1}{y(1-y)} = \frac{A - (A-B)y}{y(1-y)}$$

$$\Rightarrow \boxed{A = 1}$$

$$-(A-B) = 0$$

$$\Rightarrow \boxed{B = 1}$$